

Lab 1B — Operations, Secrets, and Incident Response

AWS Console Step-by-Step Setup Guide

Purpose

This document converts 1b_lab.md into exact AWS Console steps. You are extending the Lab 1A system with observability, alarms, and configuration storage that will later be used during an incident.

Step 1 — Store Configuration in Parameter Store

- Open AWS Console → Systems Manager → Parameter Store → Create parameter
- Create the following parameters:
- /lab/db/endpoint (String or SecureString)
- /lab/db/port
- /lab/db/name
- Verify values match the currently working database

Step 2 — Verify EC2 IAM Role Permissions

- Open IAM → Roles → EC2 application role
- Confirm permissions include:
- ssm:GetParameter / GetParameters
- secretsmanager:GetSecretValue
- logs:PutLogEvents
- Do not grant wildcard admin permissions

Step 3 — CloudWatch Logs

- Open CloudWatch → Log groups
- Confirm /aws/ec2/lab-rds-app exists
- Trigger a temporary DB failure to ensure ERROR logs appear
- Logs must clearly show DB connection failures

Step 4 — SNS Topic (PagerDuty Simulation)

- Open SNS → Topics → Create topic
- Name: lab-db-incidents
- Create Email subscription and confirm it

Step 5 — CloudWatch Alarm

- Open CloudWatch → Alarms → Create alarm
- Metric: Lab/RDSApp → DBConnectionErrors
- Threshold: ≥ 3 errors in 5 minutes
- Action: Notify lab-db-incidents SNS topic